

FAQ-23: How Do I Monitor Box With Dynatrace?

Series: FAQ — Frequently Asked Questions | **Reference:** 23 — Monitoring Box With Dynatrace | **Created:** August 2026 | **Last Updated:** 09/28/2026

Overview

Box holds the documents. That is the whole reason this integration comes up: the audit trail of who opened, downloaded, shared, and deleted them is a compliance artifact, and it lives in a system nobody in the observability team administers. The request usually arrives phrased as a logging problem — "we need Box logs in Dynatrace" — and is answered as though the only open question were which ingestion endpoint to point at.

It is not. Two things about this integration are commonly assumed the wrong way round, and both change the design:

- **There is no turnkey Dynatrace integration for Box.** No Hub listing, no extension to activate. Verified three independent ways in August 2026 — Hub catalog search, web search, and the Dynatrace documentation assistant — all empty. You are building and owning this.
- **Box cannot push its audit feed to you.** Box webhooks exist, and they are the wrong tool: they attach to individual files and folders, cap at 1,000 resources per application, cannot be set on the root folder, and never carry logins, admin actions, or security alerts. The enterprise audit feed is available *only* by polling, and Box states plainly that it "does not support long polling." Every working design is a scheduled pull that owns a cursor.

That second point is the expensive one. A team that starts from "set up a webhook" will build a receiver, wire it correctly, and then discover that the events they actually needed were never eligible to arrive. This entry exists partly to prevent that specific week.

This entry is a **worked example of FAQ-19**. The generic decisions — signal classes, route selection, extract-then-discard, bucket isolation — are made there; here they are

made concretely for Box, with a sizing example at **1,000,000 events per day** drawn from a real migration question.

A note on the DQL in this entry. Box data will not exist in your tenant until the collector runs. Every query below is therefore written to run standalone against a synthetic record using `data record(...)` — **zero bytes scanned, zero cost** — so you can validate the shape before you have ingested anything. All queries were executed against a live tenant on 08/25/2026. Swap `data record(...)` for `fetch logs | filter log.source == "box.events"` once data flows.

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Prerequisites

Requirement	Details
Box licensing	Business plan or above for the Events API. Box Shield is a separate SKU — § 11 explains what its absence removes, and it is worth confirming before scoping
Box admin access	The service account must be an enterprise admin or co-admin holding <i>"Run new reports and access existing reports."</i> Without it the API returns nothing useful — which looks identical to "no events" (§ 17)
Box application	A Box app configured for Client Credentials Grant (server-side auth, no user interaction), authorized in the Box Admin Console
Dynatrace environment	SaaS with Grail and OpenPipeline
Outbound connectivity	If collecting via Workflow, <code>api.box.com</code> must be on the JS-runtime outbound allowlist (§ 7) — a silent-failure trap
Security sign-off	This feed carries user identity, file names, and IP addresses. Decide retention and masking before building — see FAQ-19 § 4 and § 6
Related reading	FAQ-19 (the generic pattern this instantiates), FAQ-20 (the Zscaler worked example), OPIPE / OPLOGS (OpenPipeline depth), ORGNZ (buckets), WFLOW (workflows), ALERT / SLO / DASH (downstream)

1. Short Answer

Build a scheduled collector that polls the Box Events API and posts to the Dynatrace Log Ingestion API. At typical enterprise volume — including **1,000,000 events per day** — a Dynatrace Workflow is sufficient and needs no external infrastructure.

The five decisions, made:

Decision	Answer	Why
Transport	Scheduled pull from <code>GET /events</code>	Box has no push for enterprise events. Webhooks cannot carry this feed (§ 2)

Decision	Answer	Why
Stream	<code>admin_logs_streaming</code> for live; <code>admin_logs</code> for backfill	They are not interchangeable — different retention, ordering, and duplicate behaviour (§ 4)
Collector	Workflow up to ~35,000 events per run; Python beyond that or for deep backfill	The 120-second JS-runtime timeout is the only real boundary (§ 6)
Destination	Log Ingestion API v2, dedicated <code>log.source</code>	No events-per-minute cap; 10 MB / 50,000 records per request (§ 9)
Retention	Extract metrics, then discard raw by default; dedicated bucket for what you keep	~1.5 GB/day at 1M events is a real budget line (§ 5)

Start with the signal that costs nothing. If your own applications call Box, OneAgent is already tracing those calls. Query them before building anything — see § 3.

2. What Box Is — and Why There Is No Turnkey Integration

Box is a cloud content management platform — file storage, sharing, workflow, and governance — operated entirely by Box. In observability terms that single fact settles the architecture: there is no host to install OneAgent on, no process to inject, no infrastructure to discover. Everything you can learn about Box, you learn through its API.

This is the same shape as Zscaler (**FAQ-20**) and Adobe AEM as a Cloud Service (**FAQ-18**), and it is the pattern generalized in **FAQ-19**.

2.1 No Hub listing — verified, not assumed

As of August 2026 there is no Dynatrace Hub listing, extension, or first-party integration for Box. This was checked three independent ways — the Hub catalog, general web search, and the Dynatrace documentation assistant — and all three returned nothing.

Per the corpus rule that a negative claim needs a positive source: what is being reported here is the result of three catalog checks on 08/25/2026, not an inference from one empty

search. Re-verify before quoting this in a design document; Hub listings appear over time.

The practical consequence is ownership. There is no vendor-maintained connector to inherit, and no upgrade path that will fix your integration for you. That is a normal position — it is simply better decided deliberately in week one than discovered in week three.

2.2 Why webhooks are the wrong answer — and why people reach for them

Box does publish webhooks, which is exactly why this goes wrong. They are a real feature that does a different job.

Box webhooks V2	Consequence for an audit feed
Attach to specific files or folders ; cannot be set on the root folder (ID 0)	There is no account-wide subscription to create
Cap of 1,000 webhooks per application + user combination	You could watch 1,000 folders — not an enterprise
Triggers are content actions (upload, copy, move, share, delete)	No logins, no admin actions, no SHIELD_ALERT
Delivery requires a 2xx within 30 s; Box retries up to 5× over 1 hour	A receiver outage drops events with no cursor to replay from
Token expiry degrades the payload to NO_ACTIVE_SESSION	Silent partial failure

Enterprise audit events are reachable only through `GET /events` with `stream_type` set to `admin_logs` or `admin_logs_streaming`, and Box's own documentation states that **"the enterprise event feed does not support long polling."** It is pull-only.

If someone has recommended a webhook for this, they have answered a different question. Webhooks are the right tool for reacting to changes in a specific high-value folder — a contract repository, a regulated dataset. They are structurally incapable of carrying the audit trail.

Sources:

- [Webhooks \(Box Dev Docs\)](#)
- [Webhooks V2 \(Box Dev Docs\)](#) — "Box retries webhook deliveries up to 5 times over a period of 1 hour. These numbers could be subject to change."

- [Webhook limitations V2 \(Box Dev Docs\)](#)
- [Get Enterprise Events \(Box Dev Docs\)](#)
- [Box Hub catalog \(Dynatrace\)](#)
- **Catalog observation (08/25/2026):** § 2.1's no-listing claim is the result of three independent catalog checks on that date, not a single search

3. The Four Signal Classes

FAQ-19 § 2 splits any third-party SaaS into signal classes and insists you choose which questions you are answering before choosing a transport. For Box there are four, and they differ enormously in cost.

Box — Four Signal Classes

Start with the one that costs nothing. Only two of the four require a collector.

ZERO EFFORT
Your apps calling Box
Source:
OneAgent (already on)
Answers:
Latency and errors on outbound Box API calls from your own services
Query today:
`span.kind == "client"`
`server.address`

BUILD COLLECTOR
Audit events (enterprise)
Source:
GET /events (pull)
Answers:
Who did what — logins, downloads, shares, permission changes
Constraint:
No push. No webhooks. No long polling.
Cursor must persist.

FREE IF ABOVE
Shield alerts (security)
Source:
Same event stream
Answers:
Compromised accounts, anomalous downloads, malware, ransomware
Filter on:
`event_type == "SHIELD_ALERT"`
Separate licence.

LOW LIFT
Availability
Source:
Statuspage JSON + Synthetic monitor
Answers:
Is Box up (vendor view) vs. is Box up for us (your view)
Both matter — they disagree more often than you would expect.

Sequencing
Query signal 1 before building anything — if it answers "is Box hurting us", the collector may not be needed.
Signals 2 and 3 share one collector, so build it once.

3.1 Start with the signal that costs nothing

If any of your applications call the Box API, OneAgent is **already** capturing those as client spans. No Box configuration, no credentials, no collector. This is the fastest path to an answer and it is routinely skipped.

Run this before you build anything:

```
// Signal 1 – your own services calling Box. Requires no Box integration at all.
// Note the field names: server.address and url.full are the stable fields.
// http.url does NOT exist in the semantic dictionary and returns zero rows
// silently – see § 17. Same shape, different field: span.status_code values are
// LOWERCASE, so `== "ERROR"` reports a 0% failure rate for every address.
// Divide a duration by 1ms to get a number – dividing by 1000000 still yields a duration.
fetch spans, from:-24h
| filter span.kind == "client" and endsWith(server.address, "box.com")
| summarize {calls = count(),
              failures = countIf(span.status_code == "error"),
              p90_ms = percentile(duration, 90) / 1ms},
              by:{server.address}
| fieldsAdd failure_rate_pct = round(100.0 * failures / calls, decimals: 2)
| sort calls desc
```

If that returns rows, you already have Box performance telemetry and can build a dashboard today. If it returns nothing, confirm *why* before concluding your services do not call Box — an empty result here has at least three causes (no Box traffic, no OneAgent on the calling tier, or a mistyped field name), and only one of them is informative.

3.2 What each remaining signal is actually for

Signal	Answers	Does not answer
Audit events	Who accessed, shared, downloaded, or deleted what, when, from where	Whether Box was slow or erroring
Shield alerts	Which of those actions Box's own detection considers a threat	Anything Shield is not licensed to detect
Availability	Whether Box is reachable and healthy — vendor-declared and observed	Which user was affected, or what they were doing

The common mistake is expecting audit events to answer performance questions. They are an access record, not a latency record. Box does not publish per-request timing for its own service; if you need Box performance, it comes from signal 1 (your calls) or signal 4 (synthetics), never from the audit feed.

Sources: [Global field reference \(DT docs\)](#), [List user and enterprise events \(Box Dev Docs\)](#). **Tenant observation (08/25/2026):** § 3.1's field-name caution comes from executing both forms against a live

tenant — `http.url` returned zero rows while scanning 9.8M records.

4. The Events API — The Constraint That Shapes Everything

Everything enterprise-wide comes from one endpoint, `GET /events`, in one of two mutually exclusive modes. Choosing between them is the first real design decision, and most teams need both.

	<code>admin_logs_streaming</code>	<code>admin_logs</code>
Retention	2 weeks	1 year
Latency	Near real-time	Higher
Ordering	Out of chronological order	Chronological
Duplicates	Yes — dedupe on <code>event_id</code>	None
Designed for	Live monitoring and alerting	Historical query, backfill, investigation

Use streaming for the live pipeline and historical for backfill. They are not alternatives to choose between once; they are two jobs. A collector that only runs streaming has a two-week memory and no way to recover from a longer outage. A collector that only runs historical is too slow to alert on.

4.1 The constraints that shape the collector

Constraint	Value	Design consequence
No push	<i>"The enterprise event feed does not support long polling"</i>	Scheduled pull. There is no event-driven option
Page size	500 events maximum	Paging loop required; drives the sizing math in § 5
Cursor	<code>stream_position</code> in, <code>next_stream_position</code> out	Must persist between runs or you re-ingest or skip

Constraint	Value	Design consequence
Rate limit	1,000 requests/min per user; 429 returns <code>retry-after</code>	Exponential backoff. Rarely binding — see § 5
Permission	Enterprise admin/co-admin with <i>"Run new reports..."</i>	Missing it returns nothing, indistinguishable from no events

4.2 The cursor is the whole design

Everything difficult about this collector reduces to one question: *where did I get to last time?* Box hands you `next_stream_position` with every page; you hand it back on the next call. Lose it and you either replay events you already have (paying twice) or skip events you never got (a compliance gap you cannot detect after the fact).

Because `admin_logs_streaming` retains only two weeks, a lost cursor is not always recoverable. This is the single highest-consequence property of the integration and it drives § 13.

Sources:

- [Get Enterprise Events \(Box Dev Docs\)](#)
- [List user and enterprise events \(Box Dev Docs\)](#)
- [Rate Limits \(Box Dev Docs\)](#)
- [New Enterprise Event Stream API \(Box Support\)](#)

5. Sizing: A Worked Example at 1,000,000 Events/Day

This section works a real number end to end, because the volume question is usually asked about the wrong endpoint. A migrating customer asked whether a webhook could handle 1M Box events per day; the answer they received addressed only the Dynatrace ingest limits. Both halves matter, and — as it turns out — **neither is the constraint**.

5.1 Both endpoints have ample headroom

Side	Documented limit	At 1M events/day	Verdict
Dynatrace ingest	No events-per-minute cap ; 10 MB and 50,000 records per request	~11.6 events/sec	Not a constraint
Box Events API	500 events/page; 1,000 requests/min per user	$1\text{M} \div 500 = 2,000$ pages/day $\approx$ 1.4 req/min	~0.1% of budget

1,000,000 events per day is a modest load for both systems. **The bottleneck is neither endpoint — it is that no collector exists between them.** That is the part a question about "which ingestion option handles the volume" does not surface, because the ingest API is the destination, not the connection.

5.2 What that means for a Workflow

Polling every 5 minutes gives 288 runs/day and ~3,472 events per run — about **7 pages of 500**. Against a 120-second execution ceiling that is comfortable:

Quantity	Value	Basis
Runs per day (5-min schedule)	288	Schedule
Events per run	~3,472	$1\text{M} \div 288$
Pages per run	~7	500 events/page
Estimated time per run	~15 s	~1–2 s per page — an estimate, not a documented figure
Execution ceiling	120 s	JS-runtime hard limit
Approximate headroom used	~10%	Derived

Mark the page-latency assumption clearly and measure it in your own tenant. The documented numbers here — 120 s, 500 events/page, 1,000 req/min, no ingest rate cap — are verified. The per-page round-trip time is not, and it is the one input that could move the conclusion.

5.3 The cost line nobody quotes

At roughly 1.5 KB per event:

	Volume
Per day	~1.5 GB
Per month	~45 GB

That is a real DPS line item, and it arrives whether or not anyone budgeted for it. It is also the strongest argument for the extract-then-discard posture in § 9: most of those 45 GB are `PREVIEW` and `DOWNLOAD` events whose analytical value is fully captured by a counter.

Decide retention *before* you turn the collector on. Reducing retention later does not refund what you already ingested.

Sources: [Log ingestion limits \(DT docs\)](#), [Rate Limits \(Box Dev Docs\)](#), [Get Enterprise Events \(Box Dev Docs\)](#). **Derived:** § 5.1–5.3 combine the documented per-side limits with a 1M/day rate and a ~1.5 KB average event size; the per-page latency and event size are estimates to verify locally.

6. Choosing the Collector — Workflow or Python

Choosing the Collector — Workflow or Python

The 120-second JS-runtime timeout is the whole decision. Everything else is preference.

Dynatrace Workflow (JS runtime)

No external infrastructure to run or secure

Secrets in Credential Vault — never in code
Cursor via stateClient; ingest via logsClient.storeLog
Scheduling, retry and failure alerting are built in

Hard limits: 120 s run · 256 MB · 5 MB output
Must be a STANDARD workflow — automation-utils returns 404 on simple workflows

Python collector

No execution ceiling — backfill and replay at will

pip install boxsdk — v10+, Client Credentials Grant
Cursor in your own durable store
Run on K8s CronJob, Lambda, or any scheduler

You own runtime, secrets, scheduling and alerting
box-sdk-gen is the DEPRECATED package — the name suggests otherwise. Use boxsdk.

Where 1M events/day actually lands

Polling every 5 minutes = 288 runs/day → ~3,472 events per run → ~7 pages of 500.

~7 pages

remaining headroom inside the same 120 s run — roughly 63 more pages

1M events/day sits here — about 10% of the run budget

120 s ceiling

Verdict at 1M/day: use a Workflow.

Reach for Python when you need backfill beyond the two-week window, or when a single run would exceed ~35,000 events. Page latency is an estimate — measure it in your own tenant before relying on the margin.

	Workflow (JS runtime)	Python
Infrastructure	None — runs inside Dynatrace	You provision, run, patch, secure
Execution ceiling	120 s per run (hard)	None
Memory / output	256 MB / 5 MB output	Yours
Secrets	Credential Vault	Your own secret store
Cursor	<code>stateClient</code> app state	Your own durable store
Scheduling, retry, failure alerting	Built in	You build all three
Backfill over months	Impractical	Natural fit

At 1M events/day, use a Workflow. It clears the ceiling with roughly 90% headroom (§ 5.2) and removes an entire class of operational work — there is no container to keep patched, no secret to rotate outside Dynatrace, and failure alerting is already there.

Reach for Python when:

- You need to backfill more than the two-week streaming window, or replay a year of `admin_logs`.
- Sustained volume pushes a single run past roughly 35,000 events (~70 pages).
- Policy requires the collector to live in your own estate.
- You want to enrich against systems the JS runtime cannot reach.

A common and sensible split: **Python for the one-time historical backfill, Workflow for steady state.** They write to the same `log.source` and the same bucket, and the backfill runs once.

Sources: [JavaScript runtime limits \(Dynatrace Developer\)](#), [Get Enterprise Events \(Box Dev Docs\)](#).

Derived: the ~35,000-events-per-run boundary combines the documented 120 s ceiling with the 500-event page size and an estimated per-page latency — verify locally per § 5.2.

7. Implementation A — Dynatrace Workflow

The recommended path at typical volume. No infrastructure, secrets stay in the Credential Vault, and scheduling plus failure alerting come free.

7.1 Three things to set up first

Step	What	Why it matters
1. Credential Vault entry	Store the Box <code>client_id</code> / <code>client_secret</code> as a username/password credential with the AppEngine scope	Never put Box secrets in workflow code — they are readable by anyone who can view the workflow. A credential without the AppEngine scope is not readable from workflow code
2. Outbound allowlist	Add <code>api.box.com</code> to <code>builtin:dt-javascript-runtime.allowed-outbound-connections</code>	A non-allowlisted host fails silently. This is the single most common "my workflow does nothing" cause
3. Standard workflow	Create a standard workflow, not a simple one	A simple workflow cannot contain a Run JavaScript action at all. Standard workflows also bill differently — see ALERT-03 § 1

7.2 The collector

A single **Run JavaScript** action on a 5-minute schedule. It authenticates, reads the cursor, pages until drained, ingests, and writes the cursor back.

```

import { credentialVaultClient, logsClient } from "@dynatrace-sdk/client-
classic-environment-v2";
import { stateClient, isNotFound } from "@dynatrace-sdk/client-state";

const BOX_CREDENTIAL_ID = "CREDENTIALS_VAULT-XXXXXXXXXXXXXXXXX";
const BOX_ENTERPRISE_ID = "1234567";
const CURSOR_KEY      = "box-events-cursor";
const LOG_SOURCE       = "box.events";
const PAGE_LIMIT       = 500;
const MAX_PAGES        = 60;          // stay clear of the 120 s ceiling
const INGEST_BATCH     = 1000;       // well under 50,000 records / 10 MB
const MAX_EVENT_AGE_MS = 23 * 60 * 60 * 1000; // log ingest drops timestamps
> 24 h old (§ 9.1)

export default async function () {
  // --- 1. Authenticate (Client Credentials Grant) -----
  const creds = await credentialVaultClient.getCredentialsDetails({
    id: BOX_CREDENTIAL_ID,
  });

  const tokenRes = await fetch("https://api.box.com/oauth2/token", {
    method: "POST",
    headers: { "Content-Type": "application/x-www-form-urlencoded" },
    body: new URLSearchParams({
      grant_type: "client_credentials",
      client_id: creds.username,
      client_secret: creds.password,
      box_subject_type: "enterprise",
      box_subject_id: BOX_ENTERPRISE_ID,
    }).toString(),
  });

  if (!tokenRes.ok) {
    // Fail loudly. A silent auth failure looks exactly like "no events".
    throw new Error(`Box auth failed: ${tokenRes.status} ${await
tokenRes.text()}`);
  }
  const { access_token } = await tokenRes.json();

  // --- 2. Read the cursor -----
  // "now" on first run starts at the live edge. Use 0 to replay everything
  // the streaming window still holds (up to two weeks).
  let streamPosition = "now";
  try {
    const saved = await stateClient.getAppState({ key: CURSOR_KEY });
    if (saved?.value) streamPosition = saved.value;
  } catch (e) {

```

```

    // No cursor yet – first run. Any other error should surface.
    // The SDK error has no top-level `status`; use the exported type guard.
    if (!isNotFound(e)) throw e;
  }

  // --- 3. Page until drained -----
  const seen = new Set();
  const records = [];
  let pages = 0;

  while (pages < MAX_PAGES) {
    const url =
      `https://api.box.com/2.0/events?stream_type=admin_logs_streaming` +
      `&limit=${PAGE_LIMIT}&stream_position=${encodeURIComponent(streamPosition)}`;

    const res = await fetch(url, {
      headers: { Authorization: `Bearer ${accessToken}` },
    });

    if (res.status === 429) {
      // Respect Box's backoff and finish on the next scheduled run.
      break;
    }
    if (!res.ok) {
      throw new Error(`Box events failed: ${res.status} ${await res.text()}`);
    }

    const page = await res.json();
    const entries = page.entries ?? [];

    for (const evt of entries) {
      // admin_logs_streaming can repeat events – dedupe within the run.
      if (seen.has(evt.event_id)) continue;
      seen.add(evt.event_id);

      const record = {
        content: JSON.stringify(evt),
        "log.source": LOG_SOURCE,
        "box.created_at": evt.created_at,
        "box.event_id": evt.event_id,
        "box.event_type": evt.event_type,
        "box.user.login": evt.created_by?.login,
        "box.item.name": evt.source?.item_name,
        "box.ip_address": evt.ip_address,
      };

      // Log ingest DROPS a record whose timestamp is older than 24 h. For a

```

```

    // backlog (first run from 0, a long outage), leave timestamp unset so
the
    // record takes its ingest time, and query event time on box.created_at.
    if (Date.now() - Date.parse(evt.created_at) < MAX_EVENT_AGE_MS) {
        record.timestamp = evt.created_at;
    }
    records.push(record);
}

streamPosition = page.next_stream_position;
pages++;

// A short page usually means the live edge – but Box can return fewer
than
// `limit` while events remain. Stopping here only defers them to the next
// run; the cursor still points at them, so nothing is lost.
if (entries.length < PAGE_LIMIT) break;
}

// --- 4. Ingest in batches -----
for (let i = 0; i < records.length; i += INGEST_BATCH) {
    await logsClient.storeLog({
        type: "application/json; charset=utf-8",
        body: records.slice(i, i + INGEST_BATCH),
    });
}

// --- 5. Persist the cursor LAST -----
// Only after a successful ingest. Writing it earlier means a failed ingest
// silently loses every event in this batch.
await stateClient.setAppState({
    key: CURSOR_KEY,
    body: { value: String(streamPosition) },
});

return { pages, ingested: records.length, cursor: streamPosition };
}

```

7.3 Six details in that code that are load-bearing

Detail	Why
Cursor written last	If ingest fails after the cursor advances, those events are skipped by the live collector — and once they fall out of the two-week streaming window, only an <code>admin_logs</code> backfill (§ 13.3) can recover them. Write-after-success makes a failed run replay instead of lose

Detail	Why
No TTL on the cursor	<code>stateClient TTL (validUntilTime)</code> accepts <code>now+1m ... now+90d</code> . Set one and your cursor silently expires, resetting the collector. Omit it
<code>MAX_PAGES</code> guard	Bounds the run below the 120 s ceiling. A backlog drains across several runs rather than timing out forever on the first
Throwing on auth failure	An unhandled non-OK response would leave you with an empty ingest that looks identical to a quiet Box tenant
<code>isNotFound(e)</code> on the first cursor read	The SDK's error object carries the HTTP response under <code>response</code> , not a top-level <code>status</code> — a <code>e.status !== 404</code> test is always true and makes every first run throw
No timestamp on events older than ~23 h	Log ingest drops any record timestamped more than 24 h in the past (§ 9.1). Replaying a backlog with <code>timestamp: created_at</code> loses it silently; ingest-time stamping plus <code>box.created_at</code> keeps it

7.4 Alert on the collector, not just on Box

A Workflow that stops running produces no logs — and no logs looks exactly like no activity. Add a second workflow that runs the § 13 staleness query and pages someone when it goes stale. **This is not optional at a two-week retention window.**

Sources:

- [JavaScript runtime limits \(Dynatrace Developer\)](#)
- [client-state SDK \(Dynatrace Developer\)](#) — exports the `isNotFound` type guard (checked in `@dynatrace-sdk/client-state 1.11.0` typings, 09/28/2026)
- [Manage secrets \(Dynatrace Developer\)](#) — *"Ensure to use the AppEngine scope for your credentials."*
- [Create a simple workflow \(DT docs\)](#) — *"You can use all available actions to create a simple workflow except Run JavaScript"*
- [Credential vault client \(Dynatrace Developer\)](#)
- [Client Credentials Grant \(Box Dev Docs\)](#)
- [List user and enterprise events \(Box Dev Docs\)](#)
- **Derived:** § 7.3's `write-cursor-last` rule combines the documented two-week streaming retention with

the collector resuming from its stored cursor — events behind an advanced cursor are reachable again only through an `admin_logs` backfill (§ 13.3)

8. Implementation B — Python

Use Python for historical backfill, for replay, or when sustained volume exceeds what a single 120-second run can drain.

8.1 Install the right SDK — this one is easy to get backwards

```
pip install boxsdk requests
```

`boxsdk` (v10+) is the current Box Python SDK. `box-sdk-gen` is the deprecated one. The naming is actively misleading — "gen" sounds like the newer generation, and it was, briefly: it shipped as a standalone package and was deprecated on **17 September 2025** when its functionality was absorbed into the core SDK at v10.

The PyPI release history settles it: `boxsdk` 10.16.0 shipped 23 September 2026, while `box-sdk-gen` has not released since 05 September 2025.

The second half of the trap: you install `boxsdk`, but you import `box_sdk_gen`. From v10 the `boxsdk` distribution contains only the generated package — its PyPI page says v10 *"fully and exclusively replaces the old `boxsdk` package"* — so `from boxsdk import client, CCGAuth`, which every pre-v10 example uses, fails with `ModuleNotFoundError`. The collector below imports `BoxClient`, `BoxCCGAuth` and `CCGConfig` from `box_sdk_gen`.

8.2 The collector

Syntax- and import-checked against `boxsdk` 10.16.0 (09/28/2026), including the `client.events.get_events` signature; it has not been run end-to-end against a Box enterprise.

```

"""Poll the Box enterprise event stream and ship to Dynatrace.

Steady state: --stream-type admin_logs_streaming
Backfill:     --stream-type admin_logs --created-after 2026-01-01T00:00:00Z \
              --cursor-file /var/lib/box/cursor-backfill

Requires Python 3.11+ (datetime.fromisoformat accepts Box's offsets and "Z").
"""

import argparse
import json
import os
import time
from datetime import datetime, timedelta, timezone
from pathlib import Path

import requests
from box_sdk_gen import BoxCCGAuth, BoxClient, CCGConfig
from box_sdk_gen.managers.events import GetEventsStreamType

BOX_CLIENT_ID      = os.environ["BOX_CLIENT_ID"]
BOX_CLIENT_SECRET  = os.environ["BOX_CLIENT_SECRET"]
BOX_ENTERPRISE_ID  = os.environ["BOX_ENTERPRISE_ID"]

DT_ENV_URL         = os.environ["DT_ENV_URL"]          #
https://abc12345.live.dynatrace.com
DT_API_TOKEN       = os.environ["DT_API_TOKEN"]        # scope: logs.ingest

LOG_SOURCE         = "box.events"
PAGE_LIMIT         = 500
INGEST_BATCH       = 1000      # well under the 50,000-record / 10 MB request limits
MAX_EVENT_AGE      = timedelta(hours=23) # ingest drops timestamps > 24 h old
($ 9.1)

def box_client() -> BoxClient:
    """Server-side auth. No user interaction, no refresh-token juggling."""
    config = CCGConfig(
        client_id=BOX_CLIENT_ID,
        client_secret=BOX_CLIENT_SECRET,
        enterprise_id=BOX_ENTERPRISE_ID,
    )
    return BoxClient(auth=BoxCCGAuth(config=config))

def read_cursor(cursor_file: Path, default: str) -> str:
    return cursor_file.read_text().strip() if cursor_file.exists() else
default

```

```

def write_cursor(cursor_file: Path, position: str) -> None:
    """Write atomically – a torn cursor file is worse than no cursor file."""
    cursor_file.parent.mkdir(parents=True, exist_ok=True)
    tmp = cursor_file.with_suffix(".tmp")
    tmp.write_text(str(position))
    tmp.replace(cursor_file)

def to_dt_record(evt: dict) -> dict:
    """Flatten a Box event into a Dynatrace log record."""
    created_by = evt.get("created_by") or {}
    source      = evt.get("source") or {}
    created_at = evt.get("created_at")
    record = {
        "content":      json.dumps(evt, separators=(",", ":")),
        "log.source":    LOG_SOURCE,
        "box.created_at": created_at,
        "box.event_id":  evt.get("event_id"),
        "box.event_type": evt.get("event_type"),
        "box.user.login": created_by.get("login"),
        "box.item.name":  source.get("item_name"),
        "box.item.type":  source.get("type"),
        "box.ip_address": evt.get("ip_address"),
    }
    # Only stamp event time when ingest will accept it. An older event keeps
    # its ingest time as timestamp; query its real time on box.created_at.
    if created_at:
        age = datetime.now(timezone.utc) - datetime.fromisoformat(created_at)
        if age < MAX_EVENT_AGE:
            record["timestamp"] = created_at
    return record

def ingest(records: list[dict]) -> None:
    """POST to the Dynatrace Log Ingestion API, batched."""
    url = f"{DT_ENV_URL}/api/v2/logs/ingest"
    headers = {
        "Authorization": f"Api-Token {DT_API_TOKEN}",
        "Content-Type": "application/json; charset=utf-8",
    }
    for i in range(0, len(records), INGEST_BATCH):
        batch = records[i:i + INGEST_BATCH]
        for attempt in range(5):
            resp = requests.post(url, headers=headers,
                                data=json.dumps(batch), timeout=30)
            if resp.status_code == 204:

```

```

        break
    # 200 is a PARTIAL success: some records were dropped (for example
    # "... events were not ingested because of timestamp out of
correct
    # time range"). Treat it as a failure so the cursor does not
advance.

    if resp.status_code == 200:
        raise RuntimeError(f"Dynatrace partial ingest: {resp.text}")
    # 413 means the batch expanded past 10 MB (or 16 MB after
    # preprocessing) – halving is the fix, not retrying.
    if resp.status_code == 413 and len(batch) > 1:
        mid = len(batch) // 2
        ingest(batch[:mid])
        ingest(batch[mid:])
        break
    if resp.status_code in (429, 500, 502, 503, 504):
        time.sleep(2 ** attempt)
        continue
    resp.raise_for_status()
    raise RuntimeError(f"Unexpected ingest response
{resp.status_code}")
    else:
        raise RuntimeError("Dynatrace ingest failed after 5 attempts")

def collect(stream_type: str, cursor_file: Path,
            created_after: datetime | None = None) -> int:
    client = box_client()
    # "now" returns no events, only the live-edge position; "0" replays
    # everything the stream still holds. Backfill wants the latter.
    default = "0" if stream_type == "admin_logs" else "now"
    position = read_cursor(cursor_file, default)
    seen: set[str] = set()
    total = 0

    while True:
        # The SDK handles auth, retry and token refresh for us.
        page = client.events.get_events(
            stream_type=GetEventsStreamType(stream_type),
            stream_position=position,
            limit=PAGE_LIMIT,
            created_after=created_after,
        )
        entries = page.entries or []

        records = []
        for evt in entries:
            data = evt.to_dict()

```

```

        eid = data.get("event_id")
        # admin_logs_streaming repeats events; admin_logs does not.
        if eid in seen:
            continue
        seen.add(eid)
        records.append(to_dt_record(data))

    if records:
        ingest(records)
        total += len(records)

    # Persist only after a successful ingest – and on an EMPTY page too,
    # or a cursor that started at "now" is never initialized.
    position = str(page.next_stream_position)
    write_cursor(cursor_file, position)

    # Empty means the live edge. A short page may not be – Box can return
    # fewer than `limit` while events remain – but stopping only defers
    # them to the next run; the saved cursor still points at them.
    if len(entries) < PAGE_LIMIT:
        break

    return total

if __name__ == "__main__":
    ap = argparse.ArgumentParser()
    ap.add_argument("--stream-type", default="admin_logs_streaming",
                    choices=["admin_logs_streaming", "admin_logs"])
    ap.add_argument("--created-after", default=None,
                    help="RFC3339 timestamp; admin_logs backfill only")
    ap.add_argument("--cursor-file", type=Path,
                    default=Path(os.environ.get("BOX_CURSOR_FILE",
                                                  "/var/lib/box/cursor")),
                    help="use a separate file per stream type")
    args = ap.parse_args()

    after = (datetime.fromisoformat(args.created_after)
              if args.created_after else None)
    count = collect(args.stream_type, args.cursor_file, after)
    print(f"ingested {count} events")

```

8.3 Running it

Mode	Command	Notes
Steady state	<code>python box_collector.py</code>	Every 5 min via cron, K8s <code>CronJob</code> , or a systemd timer
Backfill	<code>python box_collector.py --stream-type admin_logs --created-after 2026-01-01T00:00:00Z --cursor-file /var/lib/box/cursor-backfill</code>	Run once. The separate cursor file keeps it from clobbering the live cursor. Events older than 24 h land with their ingest time as <code>timestamp</code> — query them on <code>box.created_at</code> (§ 9.1)

Keep the two cursors apart. A backfill sharing the live cursor will drag the streaming collector backwards or forwards unpredictably — the most common way this design corrupts itself.

8.4 Where to run it

Host	Fit
Kubernetes <code>CronJob</code>	Best fit if you already run K8s — secrets, scheduling, and restarts are solved
AWS Lambda / Azure Functions	Fine; keep the cursor in DynamoDB, Table Storage, or a parameter store rather than a file
A VM with cron	Works. Ensure the cursor file is on persistent storage, not ephemeral disk

Whatever runs it needs monitoring of its own — see § 13.

Sources:

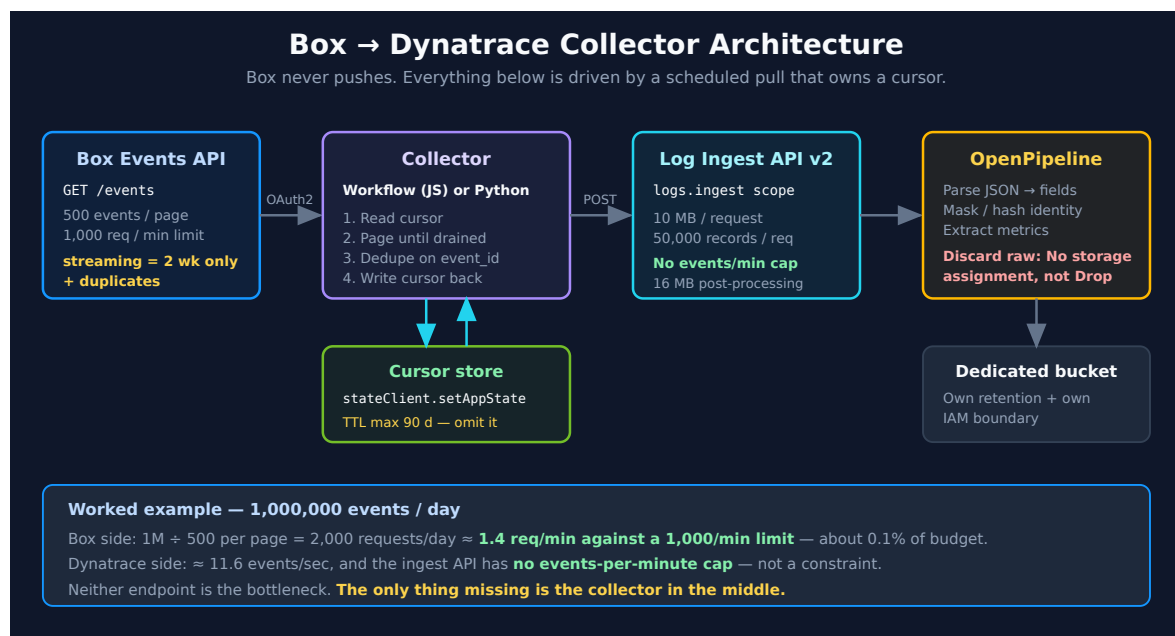
- [boxsdk on PyPI](#) — "Starting with v10, the SDK is built entirely on the generated `box_sdk_gen` package, which fully and exclusively replaces the old `boxsdk` package."
- [Deprecated Box Next Gen Python SDK \(Box Dev Docs\)](#)
- [Client Credentials Grant \(Box Dev Docs\)](#)
- [POST ingest logs \(DT docs\)](#)
- [Log ingestion limits \(DT docs\)](#) — "200 - in case some of the events in the payload have timestamps"

earlier than the current time minus 24 hours."

– List user and enterprise events (Box Dev Docs) — "now will return an empty list events and the latest stream position for initialization"; "Sometimes, the events less than the limit requested can be returned even when there may be more events remaining."

– **Derived:** § 8.1's current-vs-deprecated verdict comes from PyPI release dates (re-checked 09/28/2026) against Box's deprecation notice

9. Ingestion, Processing, and Storage



9.1 Ingest limits that actually bind

Limit	Value	Consequence
Events per minute	No limit	Volume is not the problem (§ 5)
Payload per request	10 MB	Batch by size, not only by count
Records per request	50,000	Rarely the binding limit — size hits first

Limit	Value	Consequence
Post-processing expansion	Rejected above 16 MB	A batch inside 10 MB can still fail after processing. Halve on 413
Attributes per record	500	Ample for Box events
Attribute value length	32 kB	A large <code>additional_details</code> can approach this
Log age	24 h — 72 h from SaaS 1.348 (staged tenant rollout)	A record timestamped earlier is dropped ; a mixed batch returns 200 with a "not ingested" message, not an error. Backfill must not stamp old events with their event time (§ 7.3, § 8.2)

9.2 Extract, then discard — in that order

The mechanics and the trap are in **FAQ-19 § 4**; do not configure this without reading it. In short:

Extract	Example	Retention posture
Metric	Event counts by <code>event_type</code> and user; download volume by user; failed-login counts	Keep the metric. Discard the source record unless separately justified
Business event	A file in a regulated folder was shared externally	Keep , minimal fields
Alert	<code>SHIELD_ALERT</code> (§ 11)	Keep with full context
Raw record	Every <code>PREVIEW</code> and <code>DOWNLOAD</code> event	Discard by default. This is the bulk of your 45 GB/month

The trap, stated once more because it silently defeats the whole design: `Drop record` in the **Processing** stage pre-empts every extractor downstream — your metrics never get built. Use `No storage assignment` in the **Bucket assignment** stage instead: the record still runs through metric extraction, which comes after it, and is simply not stored. Extract first, discard last.

Field-level controls in the Processing stage, before storage:

Field class	Control
<code>created_by.login</code> , <code>created_by.name</code>	Hash where the analysis only needs correlation, not identity. Note a hash of a corporate email is enumerable — a control against casual exposure, not a privacy guarantee
<code>source.item_name</code>	Consider masking. File names leak project names, client names, and deal names
<code>ip_address</code>	Keep — needed for the § 11 and § 12 access analysis
<code>additional_details</code> on high-volume event types	Remove. Cheapest control, apply first

9.3 Bucket isolation

Put Box events in a **dedicated bucket**. Two independent reasons, either sufficient:

- **Retention differs.** An audit trail is typically kept far longer than application logs, and paying application-log retention across 45 GB/month of Box events is expensive.
- **Access differs.** This feed names individuals and the documents they touched. It should be readable by security and compliance, not by everyone who can read application logs. A bucket is the boundary that makes that enforceable — see **ORGNZ**.

Sources:

- [Log ingestion limits \(DT docs\)](#) — "The event is dropped if the log event contains a timestamp before the current time minus 24 hours."
- [What's new in Dynatrace SaaS 1.348 \(DT docs\)](#) — "Log ingestion now accepts log data up to 72 hours old"; pre-release, read 09/28/2026
- [Processing in OpenPipeline \(DT docs\)](#)
- [Configure data storage and retention for logs \(DT docs\)](#) — "This means you can extract metrics and generate alerts from records that you won't store."
- [Log ingestion \(DT docs\)](#) — the ingestion-route overview this section's API choice sits inside

10. Field Mapping Starting Point

Box events are JSON, so the parse is mechanical. What matters is normalizing to names that read the same as the rest of your data.

Box field	Suggested Dynatrace field	Notes
<code>event_id</code>	<code>box.event_id</code>	Dedupe key. Required — streaming repeats events
<code>event_type</code>	<code>box.event_type</code>	Primary dimension. <code>LOGIN</code> , <code>UPLOAD</code> , <code>DOWNLOAD</code> , <code>SHARE</code> , <code>DELETE</code> , <code>SHIELD_ALERT</code> , ...
<code>created_at</code>	<code>timestamp</code>	Map to the record timestamp, not a custom field
<code>created_by.login</code>	<code>box.user.login</code>	Consider hashing (§ 9.2)
<code>created_by.id</code>	<code>box.user.id</code>	Stable across renames — prefer for joins
<code>created_by.name</code>	<code>box.user.name</code>	Display only
<code>source.type</code>	<code>box.item.type</code>	<code>file</code> , <code>folder</code> , <code>user</code> , <code>app_item</code>
<code>source.id</code>	<code>box.item.id</code>	Stable identifier
<code>source.item_name</code>	<code>box.item.name</code>	Consider masking (§ 9.2)
<code>ip_address</code>	<code>box.ip_address</code>	Retain — needed for access analysis
<code>additional_details</code>	<code>box.details.*</code>	Enterprise-stream only. Shape varies by <code>event_type</code>

Normalize before you extract. Metric dimensions and any Smartscape identifiers are computed from fields as they stand at their stage, so renaming after extraction leaves metrics carrying the old dimension names. Rename in Processing, then extract — the same ordering rule as § 9.2.

If you also ingest identity events from elsewhere (Okta, Entra ID), map `box.user.login` onto whatever principal field those feeds use. Correlating a Box download with the sign-in that preceded it is one of the highest-value things this integration enables, and it only works if the principal is spelled the same way in both.

11. Box Shield Alerts

Box Shield is Box's own threat detection. If it is licensed, its alerts arrive **in the same event stream** you are already collecting — no second integration, no extra credential. If it is not licensed, no `SHIELD_ALERT` events exist and nothing below applies.

11.1 What Shield detects

Alert type	Trigger
Suspicious Locations	Access from unusual or excluded geographies
Suspicious Sessions	Unusual user-agents, new IPs, implausible travel
Anomalous Downloads	Download behaviour consistent with data theft
Malicious Content	Malware detected on upload
Ransomware Activity	File extensions consistent with ransomware (Shield Pro only)

11.2 The alert payload

Shield alerts follow the standard event schema with `event_type == "SHIELD_ALERT"`, carrying an `additional_details.shield_alert` object:

Field	Use
<code>rule_category</code> , <code>rule_name</code>	Classification — the primary dimension for routing
<code>risk_score</code>	Numeric severity; good threshold for problem creation
<code>priority</code>	Box's own severity label
<code>alert_summary</code>	Context: IPs, affected files, timestamps
<code>link</code>	Deep link back to the Shield console — put this in the notification
<code>rule_response_action</code>	What Shield already did (e.g. terminated the session)

The `link` field deserves emphasis. An alert that reaches an analyst without a route back to the vendor console costs several minutes of navigation per incident. Carry it through processing into the notification payload.

11.3 Routing them

Shield alerts are the one Box signal that should page someone. Route them per **FAQ-21**:

1. Extract `SHIELD_ALERT` events in OpenPipeline into a **davis event** or business event.
2. Attach ownership so the trigger can route on it — FAQ-21 covers why `dt.owner` on an event needs an explicit, non-retroactive Problem fields mapping.
3. Trigger a **problem-triggered workflow**, filtering on `rule_category` and `risk_score`.
4. **Keep these records.** This is the category where raw retention is justified — § 9.2's discard-by-default posture explicitly exempts it.

```
// Shield alert extraction. Runs standalone against a synthetic record – zero
// bytes scanned. Replace the `data record(...)` line with:
//   fetch logs, from:-24h | filter log.source == "box.events"
// once your collector is running.
data record(content = "
{"event_id":"a1","event_type":"SHIELD_ALERT","created_at":"2026-08-
25T10:15:02-07:00","created_by":
{"login":"jdoe@example.com"},"additional_details":{"shield_alert":
{"rule_category":"Anomalous Download","rule_name":"Bulk
download","risk_score":78,"priority":"high","alert_id":"9911"}}})
| parse content, "JSON:evt"
| filter evt[event_type] == "SHIELD_ALERT"
| fieldsAdd box.alert.category    = evt[additional_details][shield_alert]
[rule_category],
                box.alert.risk_score = evt[additional_details][shield_alert]
[risk_score],
                box.alert.priority   = evt[additional_details][shield_alert]
[priority],
                box.user              = evt[created_by][login]
| fields box.alert.category, box.alert.risk_score, box.alert.priority,
box.user
```

Sources: [Shield Alert Events \(Box Dev Docs\)](#), [Box Shield \(Box\)](#). **Derived:** § 11.3's routing sequence applies FAQ-21's four-axis model to the Shield payload; the recommendation to carry `link` into the notification is an operational inference, not a documented requirement.

12. Querying Box Events in Grail

Each query below runs standalone at **zero bytes scanned**. Swap the `data record(...)` line for `fetch logs, from:-24h | filter log.source == "box.events"` once data flows.

12.1 Parse a raw Box event

If you ingested `content` as the raw JSON (as both collectors in § 7 and § 8 do), this is how you get at the fields at query time. Doing it in OpenPipeline instead is cheaper — see § 9.2 — but this form is useful while validating.

```
// Parse a raw Box event record into queryable fields.
// Runs standalone – zero bytes scanned.
data record(content = "{\"event_id\":\"f82c314e-9a01-4f2b-b6d2-1c7a55e0d311\",\"event_type\":\"LOGIN\",\"created_at\":\"2026-08-25T10:15:02-07:00\",\"created_by\":{\"type\":\"user\",\"id\":\"12345\",\"name\":\"Jane Doe\",\"login\":\"jdoe@example.com\"},\"source\":{\"type\":\"user\",\"id\":\"12345\"},\"ip_address\":\"203.0.113.42\"}")
| parse content, "JSON:evt"
| fieldsAdd box.event_id    = evt[event_id],
              box.event_type = evt[event_type],
              box.user       = evt[created_by][login],
              box.ip         = evt[ip_address]
| fields box.event_id, box.event_type, box.user, box.ip
```

12.2 Verify deduplication is working

`admin_logs_streaming` returns duplicates by design. At 1M events/day an unnoticed duplicate rate is a direct and recurring cost. Run this weekly.

```
// Duplicate rate on the Box feed. Anything above ~0 means dedupe is not
// working – check that the collector's `seen` set spans the whole run.
// Runs standalone against synthetic records – zero bytes scanned.
data record(box.event_id = "evt-001", box.event_type = "LOGIN"),
      record(box.event_id = "evt-001", box.event_type = "LOGIN"),
      record(box.event_id = "evt-002", box.event_type = "DOWNLOAD"),
      record(box.event_id = "evt-003", box.event_type = "SHIELD_ALERT")
| summarize {ingested = count(), unique = countDistinct(box.event_id)}
| fieldsAdd duplicates = ingested - unique
| fieldsAdd duplicate_pct = round(100.0 * duplicates / ingested, decimals: 1)
```

12.3 Confirm records landed in the right bucket

A bucket misassignment is silent — the data arrives, is queryable, and quietly inherits the wrong retention and the wrong access boundary.

```
// Which bucket are Box records actually in? Expect your dedicated bucket,  
// not `default_logs`. An empty result means no Box data yet — not success.  
fetch logs, from:-24h  
| filter log.source == "box.events"  
| summarize records = count(), by:{dt.system.bucket}  
| sort records desc
```

12.4 Daily volume against the sizing assumption

Confirm reality matches the § 5 budget. A sustained overshoot is the earliest warning that retention posture needs revisiting.

```
// Daily Box event volume vs. a 1M/day planning assumption.  
// Use interval:24h — interval:1d raises a calendar-duration warning.  
fetch logs, from:-7d  
| filter log.source == "box.events"  
| makeTimeseries events = count(), interval:24h  
| fieldsAdd daily_avg = arrayAvg(events)  
| fieldsAdd pct_of_1m_budget = round(100.0 * arrayAvg(events) / 1000000,  
decimals: 1)
```

13. Operating the Collector

This section is not optional, and it is the one most often skipped.

The failure mode: the collector stops. It produces no logs. No logs is indistinguishable from a quiet Box tenant. Nobody notices. Two weeks later the `admin_logs_streaming` window has rolled past the gap and **those events are gone from the live stream** — the only way back is an `admin_logs` backfill, which reaches back one year (§ 13.3), and past a year they exist only in the Box Admin Console's exported reports, outside the API.

Because Box retains the streaming feed for only two weeks, your detection window for a collector outage is *shorter than two weeks*. Alert on it.

13.1 The staleness query — and the bug in the obvious version

The natural way to write this check has a defect worth seeing, because it fails in the exact direction that hurts.

```
| fieldsAdd collector_status = if(minutes_since_last > 15, "STALLED", else: "OK")
```

When the collector has **never** run, `last_event` is null, so `now() - last_event` is null, `round(null)` is null, and `null > 15` evaluates false — landing on **"OK"**. The health check reports healthy precisely when there is no data at all. Verified against a live tenant on 08/25/2026: this form returned `collector_status: "OK"` alongside `events: 0`.

Test the record count explicitly:

```
// Collector health. Note the explicit events == 0 branch: without it, a
// collector that has NEVER run reports "OK", because null > 15 is false.
// Verified against a live tenant 08/25/2026.
fetch logs, from:-24h
| filter log.source == "box.events"
| summarize {last_event = takeMax(timestamp), events = count()}
| fieldsAdd minutes_since_last = round((now() - last_event) / 1m, decimals: 1)
| fieldsAdd collector_status = if(events == 0, "NO DATA – collector never ran
or is misconfigured",
                                else: if(minutes_since_last > 15, "STALLED",
                                else: "OK"))
```

13.2 What to alert on

Condition	Severity	Why
No Box events for > 15 min	High	At a 5-min schedule, three missed runs is a real outage
No Box events ever (<code>events == 0</code>)	High	Misconfiguration — usually the outbound allowlist or the admin permission
Duplicate rate rising above ~0	Medium	Dedupe broken; you are paying twice
Daily volume deviating sharply from baseline	Medium	Either a Box-side change or a partial collection failure
Workflow execution failure	High	Route via WFLOW — it is already built in

13.3 Recovering from an outage

Outage length	Recovery
< 2 weeks	Reset the cursor to <code>0</code> and re-run. The streaming window still holds the events; dedupe on <code>event_id</code> protects against the overlap. Anything older than 24 h lands with its ingest time as <code>timestamp</code> (log ingest drops older timestamps — § 9.1); query recovered events on <code>box.created_at</code>
> 2 weeks	Streaming cannot help. Backfill with <code>admin_logs</code> and <code>created_after</code> (§ 8.3) — up to one year, chronological and duplicate-free, again with event time carried in <code>box.created_at</code> rather than <code>timestamp</code>
Cursor lost, collector healthy	Same as above: pick the stream by how far back the gap goes

Recovery is complete in content, not in time: recovered events keep their real time in `box.created_at`, so time-bucketed dashboards built on `timestamp` show the backlog at the moment it was ingested. The existence of a one-year `admin_logs` backfill path is exactly why § 8's Python collector is worth having even in a Workflow-first design. **Build it before you need it** — writing a backfill tool during an active compliance gap is not when you want to be learning the API.

Sources: [Get Enterprise Events \(Box Dev Docs\)](#), [Log ingestion limits \(DT docs\)](#). **Tenant observation (08/25/2026):** § 13.1's null-comparison behaviour was observed directly by executing both query forms against a live tenant. The Box page also states the retention split behind § 13.3: *"One year of enterprise events are available when stream_type is set to admin_logs."*

14. Availability Monitoring

Two different questions, two different answers, and they disagree more often than you would expect.

Question	Source	What it tells you
Is Box up?	<code>https://status.box.com/api/v2/status.json</code>	Box's own declaration, per component
Is Box up for us?	Dynatrace Synthetic HTTP monitor	Reachability and latency from your locations and networks

14.1 The Box status feed

Box runs a Statuspage instance with a public JSON API — verified reachable and well-formed on 08/25/2026:

Endpoint	Returns
<code>/api/v2/status.json</code>	Overall indicator and description
<code>/api/v2/summary.json</code>	Per-component status — Content API, Box Web Application, Login/SSO, Uploads/Downloads, Box Platform / API, and others

`summary.json` is the more useful of the two: "Box is up" is far less actionable than "the Content API is degraded", especially when your integration depends on exactly that component.

Poll it from the same Workflow that collects events — it is one extra `fetch` and requires no authentication.

14.2 Synthetic monitoring — when it adds something

A vendor status page reports what the vendor has decided to declare, on the vendor's timeline. A synthetic monitor reports what your users would actually experience. Add synthetics when:

- Box is on a critical business path and you need independent confirmation before opening a vendor ticket.
- You need per-location or per-network detail — a problem in one office is invisible to a global status page.
- You want Box availability in an SLO (§ 15) and need a signal you control the measurement of.

Point an HTTP monitor at a stable, authenticated-free endpoint and alert on both availability and latency. Do not point synthetics at the Events API — you would be consuming your own rate-limit budget to test a service you are already polling.

Sources: [Box status page \(Box\)](#). **Observation (08/25/2026):** § 14.1's endpoint list and component names were read directly from the live Statuspage API on that date.

15. Dashboards, Alerting, and SLOs

15.1 Dashboards worth building

Dashboard	Audience	Content
Box security posture	Security	Shield alerts by category and risk score; external shares; failed logins; access from new geographies
Box usage	IT / business	Active users, uploads and downloads by volume, most-active folders, storage trend
Box integration health	Observability team	Collector freshness (§ 13), duplicate rate, daily volume vs. budget, ingest failures
Your apps ↔ Box	Application teams	Client-span latency and error rate against <code>box.com</code> (§ 3.1)

The third one is the one teams forget, and it is the one that protects the other three. A security dashboard fed by a dead collector shows an encouraging absence of alerts.

15.2 What deserves an alert

Signal	Alert?	Notes
<code>SHIELD_ALERT</code> above a risk-score threshold	Yes	Page security. Route per FAQ-21
Collector stale or absent	Yes	§ 13 — highest-consequence operational alert here

Signal	Alert?	Notes
Failed-login spike	Yes	Credential-stuffing signal
External share of a regulated folder	Yes , if you can identify those folders	Otherwise noise
Individual downloads	No	Volume without signal. Use Shield's anomaly detection
Box status page degradation	Maybe	Informational unless Box is on a critical path

15.3 SLOs

Box is a vendor service — you cannot control its reliability, so an SLO on Box itself is a reporting instrument, not an engineering target. What genuinely belongs in an SLO is **your integration**:

SLO	SLI	Why it is the right target
Audit completeness	% of 5-min windows with at least one successful collection	Directly measures the compliance guarantee you actually made
Collection freshness	% of time the newest event is < 15 min old	The detection window that keeps § 13 honest

The first of those is the one an auditor will ask about. See the **SLO** series for construction.

16. Implementation Sequence and Checklist

16.1 Sequence

Phase	Do	Gate before moving on
1. Free signal	Run the § 3.1 span query	Know whether you already have Box performance data
2. Scope	Decide which signal classes you need and why	Written answer to "which questions are we buying?"

Phase	Do	Gate before moving on
3. Box side	Create the app (CCG), authorize it, grant admin + reports permission	A manual <code>curl</code> against <code>/events</code> returns entries
4. Dynatrace side	Credential Vault entry, <code>api.box.com</code> allowlisted, bucket created	Allowlist confirmed — silent failures start here
5. Collector	Build per § 7 (or § 8), run at low frequency	Events arriving in the right bucket (§ 12.3)
6. Processing	OpenPipeline: parse, mask, extract metrics, discard raw	Metrics exist <i>and</i> raw volume has dropped
7. Operate	Staleness alert (§ 13), duplicate check, volume check	Alerts proven by deliberately stopping the collector
8. Downstream	Dashboards, Shield routing, SLOs	—

Phase 7 is proven by breaking it on purpose. An untested staleness alert is an assumption, and the cost of that assumption being wrong is a permanent, undetectable audit gap.

16.2 Administrator checklist

Box side

- ☐ Box app created with **Client Credentials Grant**
- ☐ App authorized in the Box Admin Console (a fresh app is *not* authorized by default)
- ☐ Service account is enterprise admin/co-admin with **"Run new reports and access existing reports"**
- ☐ Confirmed `GET /events?stream_type=admin_logs_streaming` returns entries with a manual call
- ☐ Shield licensing confirmed (or § 11 knowingly skipped)

Dynatrace side

- ☐ `client_id` / `client_secret` in the **Credential Vault** — not in workflow code
- ☐ `api.box.com` **on the JS-runtime outbound allowlist** (Workflow path only)
- ☐ Collector is a **standard** workflow, not simple
- ☐ Dedicated bucket created with agreed retention

- [] `logs.ingest` token scope (Python path only)
- [] OpenPipeline: parse → mask/hash → extract metrics → **No storage assignment** in the Bucket assignment stage, never `drop record` in Processing

Operations

- [] Cursor persisted, **no TTL**, written only after successful ingest
- [] Staleness alert live **and tested by stopping the collector**
- [] Duplicate-rate check scheduled
- [] Backfill path (§ 8) built and tested *before* it is needed — including that events older than 24 h arrive (ingest-time `timestamp` , real time in `box.created_at`)
- [] Retention and masking signed off by whoever owns the data

17. Common Gotchas

#	Gotcha	Symptom	Fix
1	Reaching for webhooks	Receiver works; the events you need never arrive	Webhooks are file/folder-scoped and carry no audit events. Poll <code>/events</code> (§ 2.2)
2	Host not on the outbound allowlist	Workflow succeeds, ingests nothing, logs nothing useful	Add <code>api.box.com</code> to <code>builtin:dt-javascript-runtime.allowed-outbound-connections</code>
3	Missing the reports permission	API returns 200 with no entries	Grant "Run new reports and access existing reports" to the service account
4	TTL on the cursor	Collector silently resets weeks later	<code>stateClient</code> TTL caps at 90 days. Omit <code>validUntilTime</code>
5	Cursor written before ingest	Events vanish on a failed ingest	Write the cursor only after ingest succeeds (§ 7.3)
6	Not deduplicating	Inflated counts and ingest cost	<code>admin_logs_streaming</code> repeats events. Dedupe on <code>event_id</code> (§ 12.2)

#	Gotcha	Symptom	Fix
7	<code>http.url</code> in span queries	Zero rows, reads as "no Box traffic"	The field does not exist. Use <code>server.address / url.full</code> (§ 3.1)
8	Staleness check returns "OK" with no data	Dead collector reports healthy	<code>null > 15</code> is false. Test <code>events == 0</code> explicitly (§ 13.1)
9	Drop record in the Processing stage	Raw discarded <i>and</i> metrics never built	Use <code>No storage assignment</code> in the Bucket assignment stage (§ 9.2)
10	<code>box-sdk-gen</code> instead of <code>boxsdk</code> — or from <code>boxsdk</code> <code>import ...</code>	Building on a deprecated SDK, or <code>ModuleNotFoundError</code> on v10	Install <code>boxsdk v10+</code> , import from <code>box_sdk_gen</code> (§ 8.1)
11	Simple workflow	No Run JavaScript action is available	Use a standard workflow (§ 7.1)
12	Backfill sharing the live cursor	Live collector jumps backwards or forwards	Separate cursor per stream type (§ 8.3)
13	Renaming fields after extraction	Metrics carry stale dimension names	Normalize in Processing, then extract (§ 10)
14	413 on ingest	Batch inside 10 MB still rejected	Payload expands during processing; ceiling is 16 MB. Halve the batch (§ 9.1)
15	<code>interval:1d</code>	Calendar-duration warning	Use <code>interval:24h</code> (§ 12.4)
16	Backfill stamped with event time	Replay "succeeds"; events older than 24 h are silently dropped (Python sees a 200 , not an error)	Omit <code>timestamp</code> for old events; carry <code>box.created_at</code> (§ 9.1)
17	Testing <code>e.status</code> on a client-state error	First Workflow run always throws	Use <code>isNotFound(e)</code> from <code>@dynatrace-sdk/client-state</code> (§ 7.3)

Gotchas 1, 7, and 8 share a shape worth naming: **each produces a confident, plausible, wrong answer rather than an error**. A webhook that delivers nothing, a span

query that returns zero, and a health check that says OK all look like success. Every one of them was reproduced during the research for this entry.

18. Recommended Approach

Query your own spans first. If your services call Box, that data exists now and costs nothing (§ 3.1). It may answer the question that prompted the project.

Then build one collector, in a Workflow. At 1M events/day a Workflow clears the 120-second ceiling with roughly 90% headroom, keeps secrets in the Credential Vault, and inherits scheduling and failure alerting. Poll `admin_logs_streaming` every 5 minutes, dedupe on `event_id`, persist the cursor with `stateClient` and no TTL, and write that cursor only after a successful ingest.

Write the Python backfill before you need it. It is the only recovery path for an outage longer than two weeks, and the worst time to write it is during one. It recovers the events, not their place on the timeline: log ingest drops timestamps older than 24 h, so recovered events carry their real time in `box.created_at`.

Decide retention before switching on. At ~45 GB/month, extract-then-discard is not an optimization — it is the difference between a proportionate cost and an unpleasant surprise. Keep Shield alerts and regulated-folder activity; discard the `PREVIEW` and `DOWNLOAD` bulk once a counter has captured it.

Alert on the collector, and prove the alert works by stopping it. Everything else in this entry protects data you have. This is the only control that protects data you have not collected yet — and Box's two-week streaming window means an unnoticed outage becomes permanent.

What "done" looks like: Shield alerts routed to security with a working console link; a security dashboard whose freshness you can vouch for; audit records in a bucket with retention and access someone signed; and a staleness alert that has been tested by breaking it on purpose.

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